

Django 3.1

- Django 3.1 supports Python 3.6, 3.7, and 3.8.
- Asynchronous views and middleware support
- JSONField for all supported database backends
- DEFAULT_HASHING_ALGORITHM settings
- Django 3.1 supports MariaDB 10.2 and higher.
- The admin no longer supports the legacy Internet Explorer browser.
- The admin now has a sidebar on larger screens for easier navigation.
- The settings.py generated by the startproject command now uses pathlib.Path instead of os.path for building filesystem paths.

Features Deprecated in 3.1

- `PASSWORD_RESET_TIMEOUT_DAYS` setting is deprecated in favor of `PASSWORD_RESET_TIMEOUT`.
- The `HttpRequest.is_ajax()` method is deprecated.
- `providing_args` argument for `Signal` is deprecated.
- The passing of URL kwargs directly to the context by `TemplateView` is deprecated. Reference them in the template with `view.kwargs` instead.
- The `NullBooleanField` model field is deprecated in favor of `BooleanField(null=True)`.

Asynchronous Views and Middleware

```
# views.py
async def my_view(request):
    await asyncio.sleep(0.5)
    return HttpResponse('Hello, async world!')
```

All asynchronous features are supported whether you are running under WSGI or ASGI mode.

If you make External HTTP calls from a view, You can use async View.

The ORM, cache layer, and several other parts of code that involve long-running network calls do not support async yet.

Pathlib

The settings.py generated by the startproject command now uses pathlib.Path instead of os.path for building filesystem paths.

```
# settings.py
```

```
import os
```

```
BASE_DIR = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
```

```
DATABASES = {
```

```
    'default': {
```

```
        'ENGINE': 'django.db.backends.sqlite3',
```

```
        'NAME': os.path.join(BASE_DIR, 'db.sqlite3'),
```

```
    }
```

```
}
```

```
BASE_DIR = Path(__file__).resolve(strict=True).parent.parent
```

```
'NAME': BASE_DIR / 'db.sqlite3',
```

The admin now has a sidebar

- The admin now has a sidebar on larger screens for easier navigation.

PASSWORD_RESET_TIMEOUT

- `PASSWORD_RESET_TIMEOUT_DAYS = 3`
- `PASSWORD_RESET_TIMEOUT = 5000`

Templates inside Project

settings.py

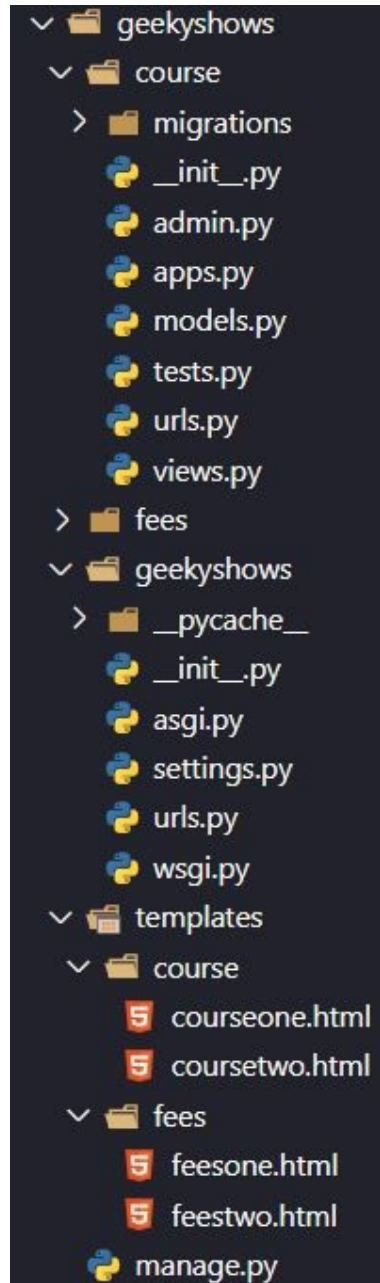
Old Django Version 3.0

```
TEMPLATES_DIR = os.path.join(BASE_DIR, 'templates')
```

```
TEMPLATES_DIR = BASE_DIR / 'templates'
```

```
INSTALLED_APPS = [  
    'course',  
    'fees'  
]
```

```
TEMPLATES = [  
    {  
        'DIRS': [TEMPLATES_DIR],  
        'APP_DIRS': True,  
    }  
]
```



Static inside Project

settings.py

```
TEMPLATES_DIR = BASE_DIR / 'templates'
```

Old Django Version 3.0

```
STATIC_DIR = os.path.join(BASE_DIR, 'static')
```

```
STATIC_DIR = BASE_DIR / 'static'
```

```
INSTALLED_APPS = [  
    'course'  
]
```

```
TEMPLATES = [  
    {  
        'DIRS': [TEMPLATES_DIR]  
    }  
]
```

```
STATIC_URL = '/static/'
```

```
STATICFILES_DIRS = [ STATIC_DIR ]
```

